DATA WAREHOUSE PROJECT

Integrated Decision Support System for Road Safety (UK STATS19)

CONTEXT

- *Road accidents have major human, economic, and social impacts.*
- *The UK STATS19 dataset provides detailed accident, vehicle, and casualty data.*
- *Centralizing and analyzing data helps identify high-risk areas and periods.*
- *Insights support better decision-making and road safety improvements.*

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PROBLEM DEFINITION

How can historical road accident data be leveraged to build a decision support system capable of identifying risk factors, tracking key indicators, and anticipating accident-prone situations?

OBJECTIVES

OBJECTIVE 1

Improving road infrastructure

OBJECTIVE 2

Dynamic deployment of police and emergency services

OBJECTIVE 3

Protecting vulnerable road users and regulating high-risk vehicles

STRATEGIC OBJECTIVES

OBJECTIVE 1

Improve road infrastructure and anticipate high-risk areas

1.1

Identify and map accident-prone areas (blackspots).

- BI / Analytics: mapping, heatmaps, accident density and frequency by area or road segment.
- ML: automatic detection of critical areas using clustering or risk-level classification.
- ML: predict the number/rate of accidents.

1.2

Analyze road-related factors associated with severe accidents: intersections, lighting, and road surface condition.

- Analytics: assessment of bivariate correlations between road surface condition / lighting and accident severity .

Improve road infrastructure and anticipate high-risk areas

1.3

Prioritize areas requiring infrastructure investment (CAPEX).

- ML: estimate the potential safety gain and prioritize areas.

OBJECTIVE 2

Dynamic deployment of police and emergency services

Analyze the spatio-temporal distribution of accidents.

- BI / Analytics: accidents by hour, day, month, season, and area.
- ML: classification of the severity of accidents level Low / Medium / High.

OBJECTIVE 3

Protecting vulnerable road users and regulating high-risk vehicles

3.1

Analyze accidents involving vulnerable road users

- BI / Analytics: casualties by age, road user type, journey purpose, imd , and severity.
- ML: Clustering of high-risk profiles .

3.2

Study accidents involving high-risk vehicles, notably heavy goods vehicles (HGVs) and delivery vehicles.

- BI / Analytics: involvement and severity rates by vehicle type and road type.

Protecting vulnerable road users and regulating high-risk vehicles

3.3

Identify vehicles associated with accident severity

- BI / Analytics: statistical analysis of severity by vehicle.

3.4

Evaluate risks related to speed and new mobility modes

- BI / Analytics: visualization of risk by speed limit and mobility type.

DECISION SUPPORT ARCHITECTURE

DECISION SUPPORT ARCHITECTURE

1. DATA SOURCES

2. PREPARATION

3. ETL

4. STORAGE

5. ANALYSIS

6. DECISION

UK STATS 19

ACCIDENTS
Date, location, severity, conditions, road condition...

VEHICLES
Vehicle type, characteristics, circumstances...

CASUALITIES
Age, sex, road user type, injury severity...

Staging area EL

Raw data, cleaning, transformation, quality control

ETL PROCESS,

Extract → **Transform** → **Load**

DATAWAREHOUSE

Dimensional model: fact table + dimensions.

POWER BI

KPIs, dashboards, and multidimensional analysis

MACHINE LEARNING

Predictive analytics (high-risk area detection, forecasting, severity...)

Decision Support

Decision-makers: authorities, police, emergency services, planners, etc.

COMPARATIVE STUDY OF METHODOLOGIES

COMPARATIVE STUDY OF DECISION SUPPORT METHODOLOGIES

Analysis Criterion	GIMSI Method (Alain Fernandez)	KIMBALL Method (Ralph Kimball)
Main Objective	Help managers make decisions.	Build and implement the decision support system.
Approach	Start from the needs of managers and the field.	Start from the requirements, then build the data and the tools.
Key Structure	KPIs, dashboards, and indicators.	Fact tables, dimensions, and ETL processes.
Orientation	Decision-makers, managers, and field users.	Technical team and end users.
Verdict	Selected for steering. Addresses public policy trade-offs.	Selected for execution: structures the development, testing, and production release

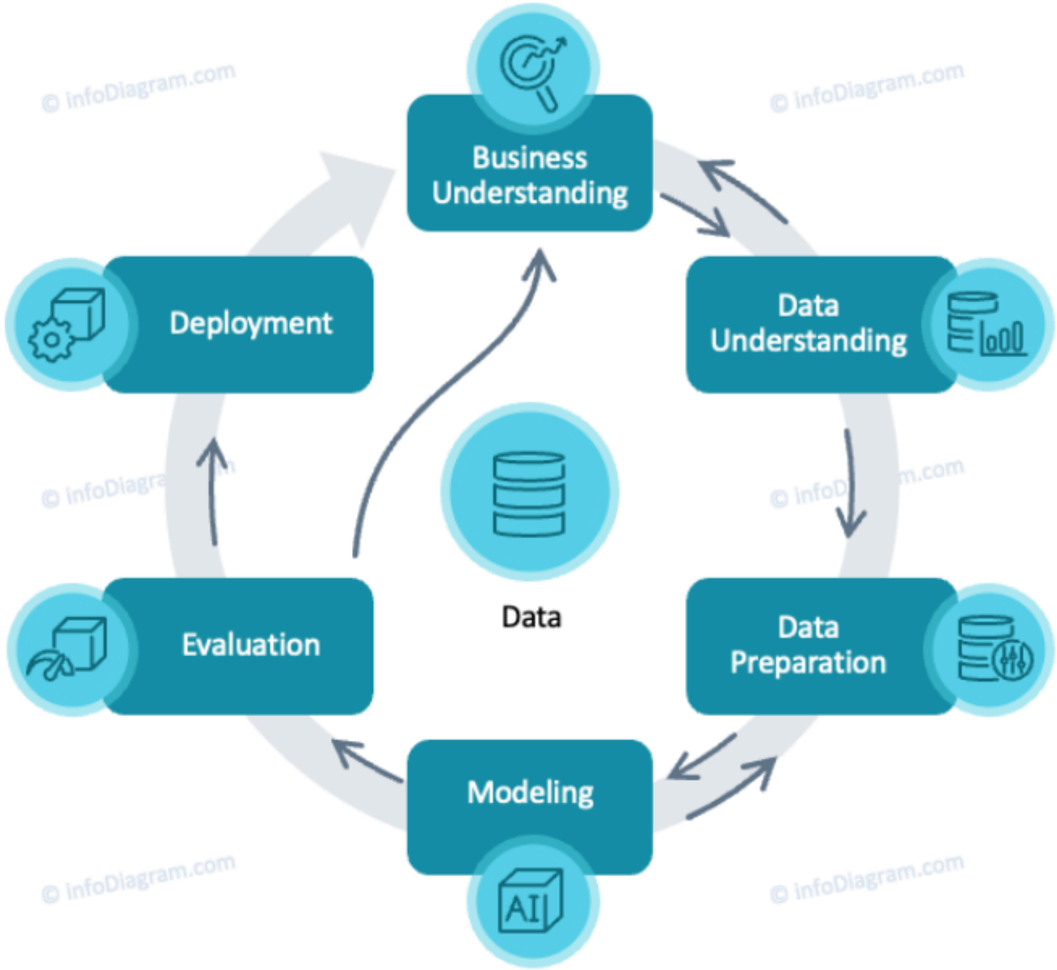
The choice

Method: GIMSI

GIMSI

We chose the GIMSI method because it starts directly from field needs to give decision-makers concrete tools, such as dashboards and KPIs.

MACHINE LEARNING METHODOLOGY



WHY CRISP-DM?

Business-driven — starts from road-safety objectives
Data-focused — supports data understanding and preparation
Flexible — suitable for classification, clustering and prediction
Iterative — allows continuous refinement and evaluation

COMPARATIVE STUDY OF MODELING APPROACHES

COMPARATIVE STUDY OF MODELING

Analysis Criterion	Inmon Approach	KIMBALL Approach.
Main Objective	Create a global and centralized enterprise data warehouse ("Top-Down" approach).	Create decentralized Data Marts based on business processes ("Bottom-Up" approach).
Architecture / Modeling	Highly normalized relational model (typically 3NF).	Dimensional model (star schema or snowflake schema).
Deployment Time	Long and complex (requires enterprise-wide design upfront).	Fast and iterative (incremental deployment by Data Mart).
Orientation	Data-driven, aiming for a "single version of the truth".	User/Business-driven, focused on analytical query performance.

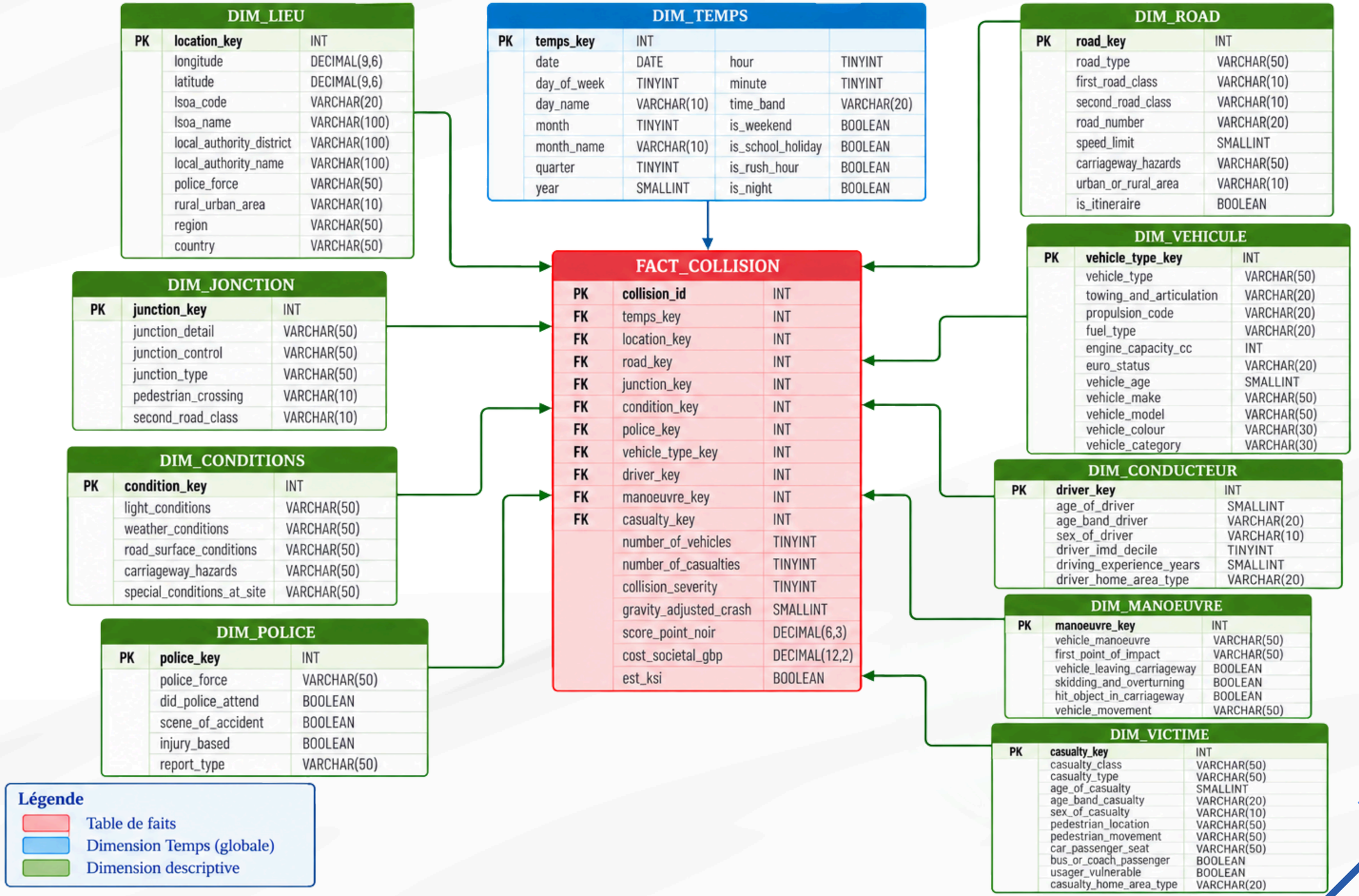
The choice

Approach: KIMBALL

Kimball for modeling

Structure the data with a dimensional model suited to BI analysis

DIMENSIONAL MODELING


DIM_LIEU
PK location_key INT
longitude DECIMAL(9,6)
latitude DECIMAL(9,6)
lsoa_code VARCHAR(20)
lsoa_name VARCHAR(100)
local_authority_district VARCHAR(100)
local_authority_name VARCHAR(100)
police_force VARCHAR(50)
rural_urban_area VARCHAR(10)
region VARCHAR(50)
country VARCHAR(50)
DIM_TEMPS
PK temps_key INT
date DATE
hour TINYINT
day_of_week TINYINT
minute TINYINT
day_name VARCHAR(10)
time_band VARCHAR(20)
month TINYINT
is_weekend BOOLEAN
month_name VARCHAR(10)
is_school_holiday BOOLEAN
quarter TINYINT
is_rush_hour BOOLEAN
year SMALLINT
is_night BOOLEAN
DIM_ROAD
PK road_key INT
road_type VARCHAR(50)
first_road_class VARCHAR(10)
second_road_class VARCHAR(10)
road_number VARCHAR(20)
speed_limit SMALLINT
carriageway_hazards VARCHAR(50)
urban_or_rural_area VARCHAR(10)
is_itineraire BOOLEAN
DIM_VEHICULE
PK vehicle_type_key INT
vehicle_type VARCHAR(50)
towing_and_articulation VARCHAR(20)
propulsion_code VARCHAR(20)
fuel_type VARCHAR(20)
engine_capacity_cc INT
euro_status VARCHAR(20)
vehicle_age SMALLINT
vehicle_make VARCHAR(50)
vehicle_model VARCHAR(50)
vehicle_colour VARCHAR(30)
vehicle_category VARCHAR(30)
DIM_JONCTION
PK junction_key INT
junction_detail VARCHAR(50)
junction_control VARCHAR(50)
junction_type VARCHAR(50)
pedestrian_crossing VARCHAR(10)
second_road_class VARCHAR(10)
FACT_COLLISION
PK collision_id INT
FK temps_key INT
FK location_key INT
FK road_key INT
FK junction_key INT
FK condition_key INT
FK police_key INT
FK vehicle_type_key INT
FK driver_key INT
FK manoeuvre_key INT
FK casualty_key INT
number_of_vehicles TINYINT
number_of_casualties TINYINT
collision_severity TINYINT
gravity_adjusted_crash SMALLINT
score_point_noir DECIMAL(6,3)
cost_societal_gbp DECIMAL(12,2)
est_ksi BOOLEAN
DIM_CONDUCTEUR
PK driver_key INT
age_of_driver SMALLINT
age_band_driver VARCHAR(20)
sex_of_driver VARCHAR(10)
driver_imd_decile TINYINT
driving_experience_years SMALLINT
driver_home_area_type VARCHAR(20)
DIM_CONDITIONS
PK condition_key INT
light_conditions VARCHAR(50)
weather_conditions VARCHAR(50)
road_surface_conditions VARCHAR(50)
carriageway_hazards VARCHAR(50)
special_conditions_at_site VARCHAR(50)
DIM_MANOEUVRE
PK manoeuvre_key INT
vehicle_manoeuvre VARCHAR(50)
first_point_of_impact VARCHAR(50)
vehicle_leaving_carriageway BOOLEAN
skidding_and_overturning BOOLEAN
hit_object_in_carriageway BOOLEAN
vehicle_movement VARCHAR(50)
DIM_POLICE
PK police_key INT
police_force VARCHAR(50)
did_police_attend BOOLEAN
scene_of_accident BOOLEAN
injury_based BOOLEAN
report_type VARCHAR(50)
DIM_VICTIME
PK casualty_key INT
casualty_class VARCHAR(50)
casualty_type VARCHAR(50)
age_of_casualty SMALLINT
age_band_casualty VARCHAR(20)
sex_of_casualty VARCHAR(10)
pedestrian_location VARCHAR(50)
pedestrian_movement VARCHAR(50)
car_passenger_seat VARCHAR(50)
bus_or_coach_passenger BOOLEAN
usager_vulnerable BOOLEAN
casualty_home_area_type VARCHAR(20)
Légende
Table de faits
Dimension Temps (globale)
Dimension descriptive

KPI’s

OBJECTIVE 1 — Road Infrastructure & High-Risk Areas

		Threshold
• Number of Collisions by Area	$N_{\text{collisions in area}}$	10 collisions/month
• Severe Collision Rate	$\frac{\text{Fatal} + \text{Serious Collisions}}{\text{Total Collisions}} \times 100$	20%
• Road Condition Severity Rate	$\frac{\text{Severe Collisions in Condition}}{\text{All Collisions in Condition}} \times 100$	25%

OBJECTIVE 2 — Police & Emergency Deployment

		Threshold
• Peak-Hour Collision Share	$\frac{\text{Collisions in Selected Hour}}{\text{Total Collisions}} \times 100$	35%
• Average Casualties per Collision	$\frac{\text{Total Casualties}}{\text{Total Collisions}}$	0.2
• Fatal Collision Rate	$\frac{\text{Fatal Collisions}}{\text{Total Collisions}} \times 100$	2%

OBJECTIVE 3 — Vulnerable Users & High-Risk Vehicles

		Threshold
• Vulnerable Road User Casualty Share	$\frac{\text{Vulnerable User Casualties}}{\text{Total Casualties}} \times 100$	35%
• HGV Collision Involvement Rate	$\frac{\text{Collisions Involving HGVs}}{\text{Total Collisions}} \times 100$	50%
• Serious or Fatal Casualty Rate	$\frac{\text{Fatal} + \text{Serious Casualties}}{\text{Total Casualties}} \times 100$	50%

CONCLUSION

To conclude, our project uses data to better understand road accidents and identify high-risk areas.

With Data Warehouse, Power BI and Machine Learning, we can analyze accidents, identify risks, predict future situations and help decision-makers use resources more effectively.

Our main goal is to use data to make roads safer and support better decisions.

Thank you for your attention.

THANKS FOR YOUR ATTENTION